MIL-E-1/45E
6 December 1971
SUPERSEDING
MIL-E-1/45D
15 February 1966

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6AG7Y

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, power amplifier

Outline --- 8-6 (EIA)

(E) Base --- B8-21 (low-loss phenolic)

Envelope --- MT8

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	sh, g3	h	nc	g1	k	g2	h	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Pg2	Pp	Ebk	Ik	(E) Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	W	W	v	mAdc	ft
Maximum:	6.9	330	---	330	0	1.5	9.0	100	95	Note 1
Minimum:	5.7	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	300	-3	150	0	---	---	---	---	---

GENERAL:

Qualification - Required

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection								
(E) 1216	Base material insulating quality	-	Zone 11, minimum	---	---	---	---	---	---
	Quality conformance inspection, part 1								
1266	Total grid current	2		0.65	II	Ic1	0	-2.0	µAdc
1256	Electrode current (1) (anode)	-		0.65	II	Ib	20	40	mAdc
1341	Power output	3	Esig = 2.1 Vac; Rp = 10,000 ohms	0.65	II	Po	2.4	---	W
1246	Audio frequency noise	-	Esig = 500 mVac; Rp = 2,000 ohms	0.65	II	EB	---	17	vu
1231	Emission	2	Eb = Ec1 = Ec2 = 20 Vdc	0.65	II	Is	180	---	mAdc
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	Rp = 2,000 ohms	---	---	Ep	---	1,000	mVac
1211	Insulation of electrodes	-		4.0	S3	---	---	---	---
1301	Heater current	-		---	---	If	610	690	mA
1336	Heater-cathode leakage	-		---	---	Ihk	---	40	µAdc
1256	Electrode current (2) (anode)	-	Eb = 150 Vdc; Ec1 = -20 Vdc	---	---	Ib	---	100	µAdc
1256	Electrode current (screen)	-		---	---	Ic2	4.0	9.0	mAdc
1306	Transconductance	-		---	---	Gm	9,200	14,200	µmhos
1331	Direct-interelectrode capacitance	-		6.5	Code E	Cg1p	---	0.060	pF
						Cin	11.5	14.5	pF
						Cout	6.5	8.5	pF
1101	Secureness of base, cap, or insert	-		---	---	---	---	---	---
1111	Base pin solder depth	-		---	---	---	---	---	---
(E) 1105	Permanence of marking	-		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 3								
1501	Intermittent life	-	Group A; Ebk = 100 V	---	---	---	---	---	---
---	Intermittent life-test end point (500 hours):								
1341	Power output	-		---	---	Po	1.6	---	W

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test to be performed at the conclusion of the holding period.
3. Place a 300-ohm resistor in series with grid leads as close to the grid as possible.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2592)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-E-1/45E

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-28384

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